Chapter (1)

- Basics
- Connection
- Ohm's Law for closed Circuit
- Kirchoff

Part (1) — Part (2)

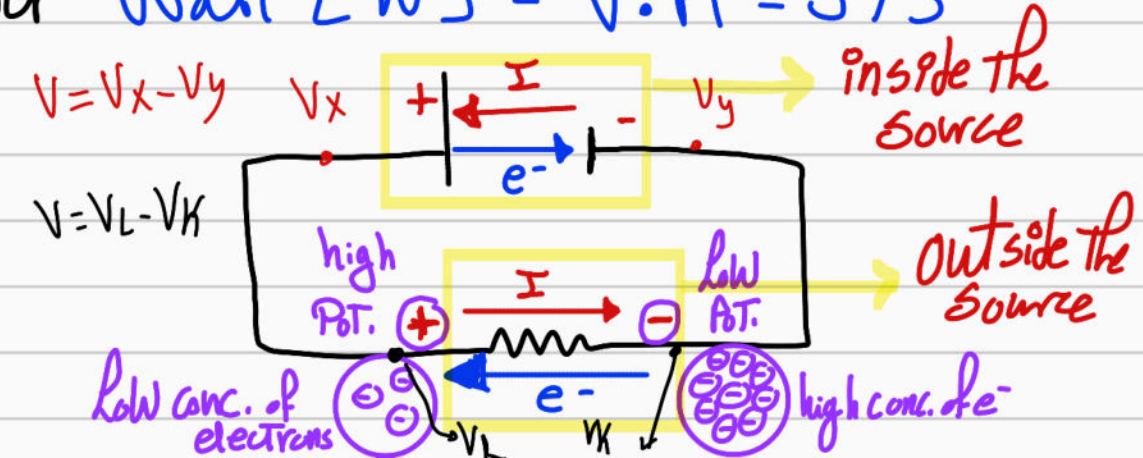
[1] Basics:

$$I = \frac{Q}{t} = \frac{V}{R} = \frac{P_w}{V}$$

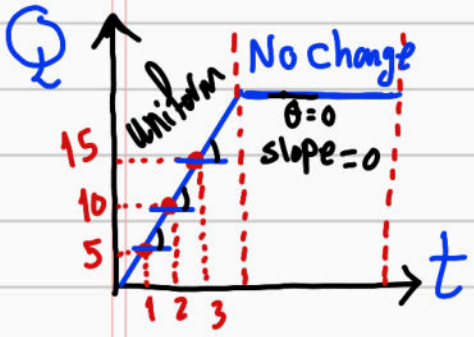
1. I → Electric Current intensity — Ampere $(A) \equiv C/S \equiv V/\Omega \equiv W/V$
2. Q → Quanitiy of electric charge — Coloumb $[C] \equiv A.S \equiv J/V$
3. V → Potential diffence between Two points — Volt $[V] \equiv A.\Omega$
4. V_B → Electromotive force — Volt $\equiv J/C$
5. R → Resistance — ohm $[\Omega] \equiv V/A \equiv W/A^2$
6. ρ_e → Resistivity — $\Omega.m \equiv V.m/A$
7. σ → Conductivity — $\Omega^{-1}.m^{-1} \equiv \frac{1}{\Omega.m} \equiv \frac{A}{V.m}$ (Simon $= \frac{1}{\Omega}$)
8. W → Work / Energy — Joule $[J] \equiv W.S \equiv V.A.S$
9. P_w → Power — Watt $[W] \equiv V.A \equiv J/S$

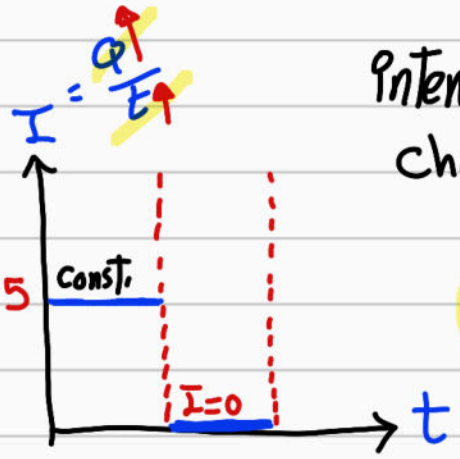
*Direction:

Conventional "Traditional"



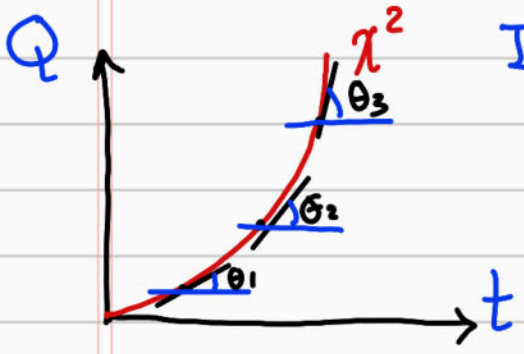
* Graphs:

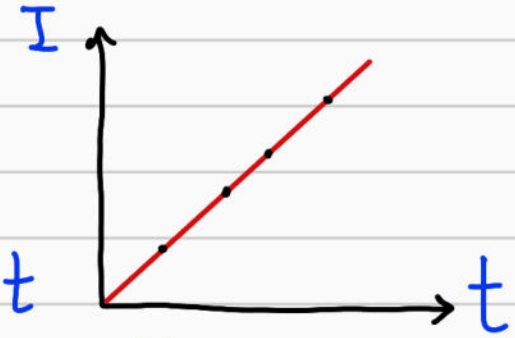




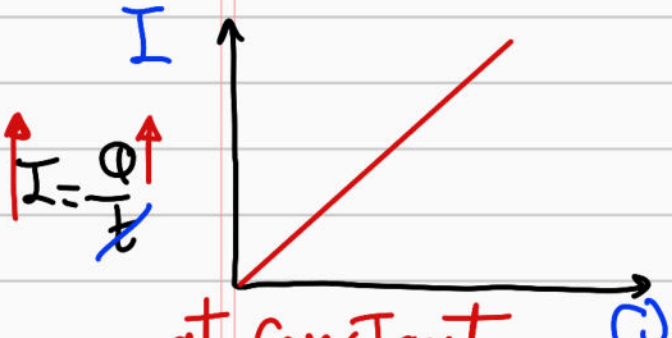
Intensity is the rate of change of Quantity

↑↑ $y = \frac{x\uparrow}{K\downarrow}$ $y = \frac{x\uparrow}{K\uparrow}$

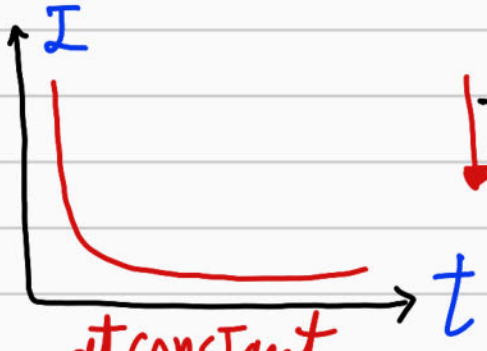




بيزيد بمعدل



at constant Time



at constant Quantity

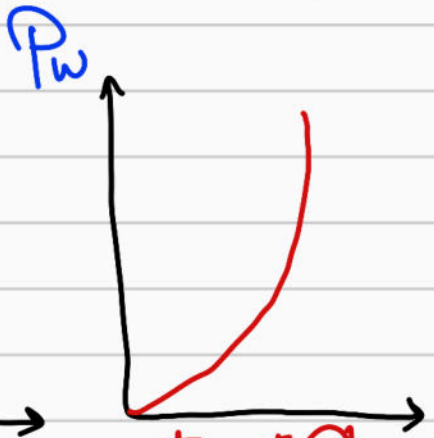
$I = \frac{Q}{t\uparrow}$



at const. (V) I

$P_W = V \times I$

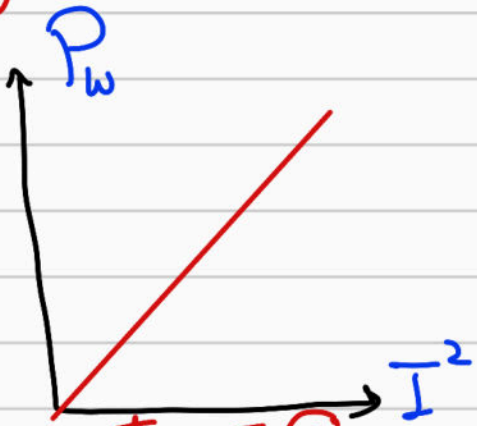
↑ $P_W \propto I$ ↑



at const. (R) I

$P_W = I^2 R$

↑↑ $P_W \propto I^2$ ↑

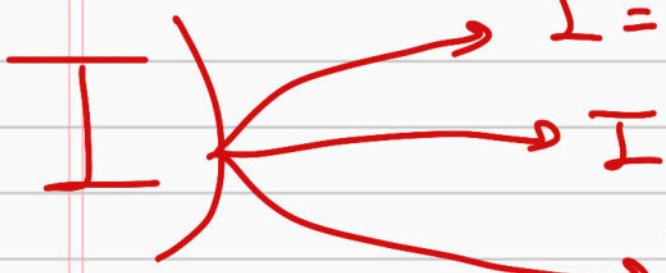


at const. (R)

$P_W = I^2 R$

↑ $P_W \propto I^2$ ↑

Out of the Box



$I = \frac{Q}{T_{per}}$

$I = Q \times freq$

$I = \frac{Q \times V}{2\pi r}$

Rotation

Ratio:

$$I = \frac{Q \cdot V}{2\pi r} \longrightarrow \boxed{\frac{I_1}{I_2} = \frac{Q_1}{Q_2} \times \frac{V_1}{V_2} \times \frac{r_2}{r_1}}$$

original

inc. To Triple $x_1 = 100 \quad x_2 = 300$

inc. by Triple $x_1 = 100 \quad x_2 = 400$ $\frac{300}{100}$

inc. To 120% $x_1 = 100 \xrightarrow{\times 120\%} x_2 = 120$

$\times 2.2$

inc. by 120% $x_1 = 100 \xrightarrow[\times 220\%]{\times (1+120\%)} x_2 = 220$

dec. To third $x_1 = 300 \xrightarrow{\times \frac{1}{3}} x_2 = 100$

dec. by Third $x_1 = 300 \xrightarrow{\times \frac{2}{3}} x_2 = 100$

dec. by 20% $x_1 = 100 \quad x_2 = 80$

dec. To 20% $x_1 = 100 \quad x_2 = 20$

Resistance [R]

$$R = \frac{V}{I} = \frac{\rho_e \times L}{A} = \frac{L}{\sigma \times A}$$

$$\frac{R_1}{R_2} = \frac{\rho_{e_1}}{\rho_{e_2}} \times \frac{L_1}{L_2} \times \frac{A_2}{A_1}$$

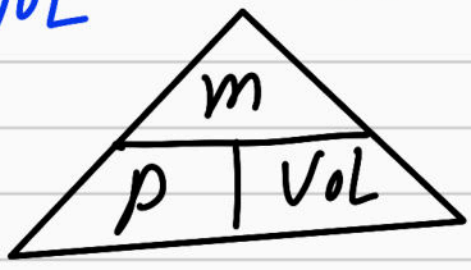
$$\frac{A_2}{A_1} = \left[\frac{r_2}{r_1}\right]^2 = \left[\frac{D_2}{D_1}\right]^2$$

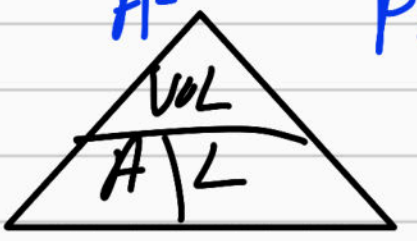
Reshape* / sTrech* / drawn* / Melted

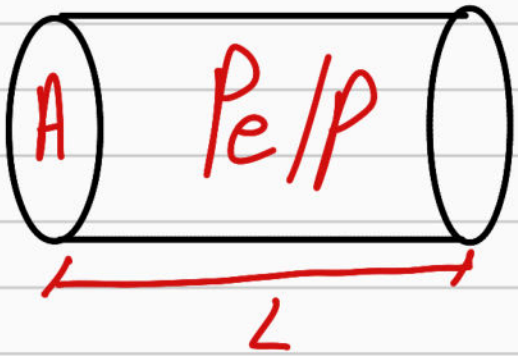
$$\frac{R_1}{R_2} = \left(\frac{L_1}{L_2}\right)^2 = \left(\frac{A_2}{A_1}\right)^2 = \left(\frac{r_2}{r_1}\right)^4 = \left(\frac{D_2}{D_1}\right)^4$$

mass / density / Volume:

$$R = \frac{\rho_e \times L^2}{VoL} = \frac{\rho_e \times L^2 \times \rho}{m} = \frac{\rho_e \times VoL}{A^2} = \frac{\rho_e \times m}{\rho \times A^2}$$

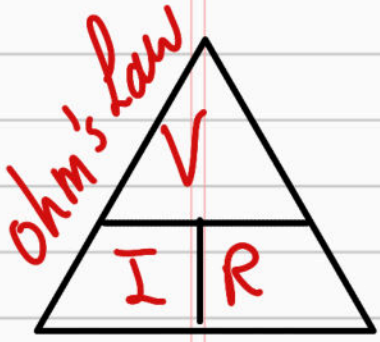


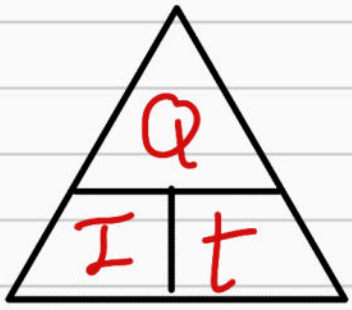


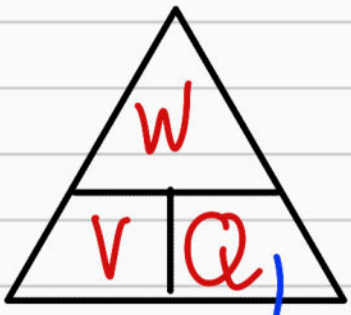


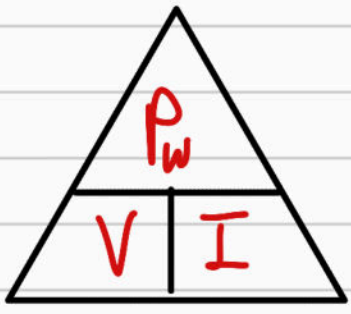
Rules Summary

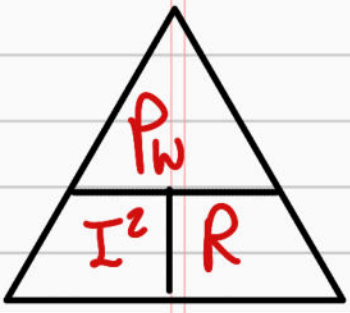
Magic Triangles



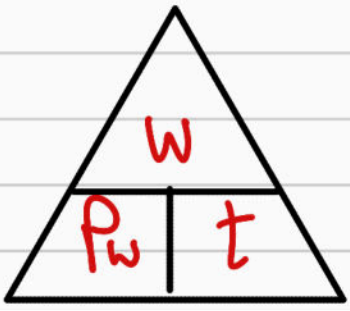


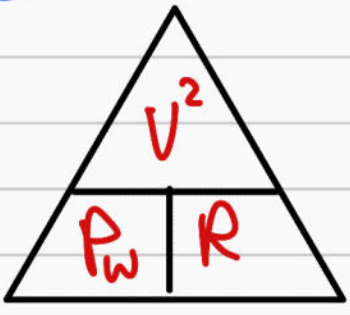


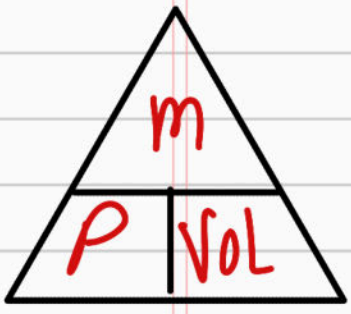


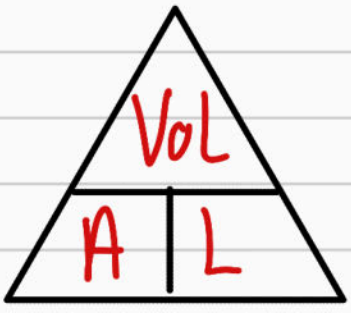


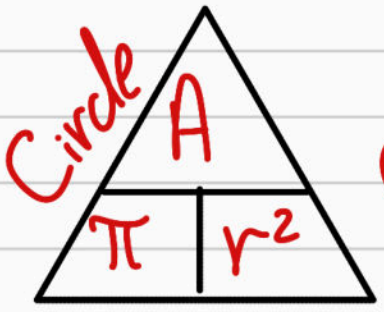


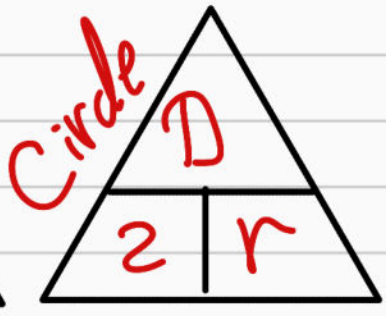


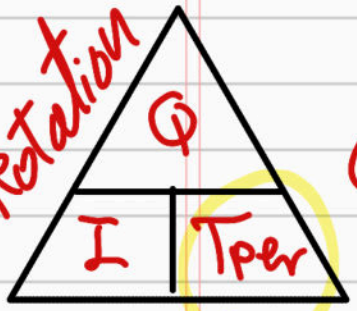


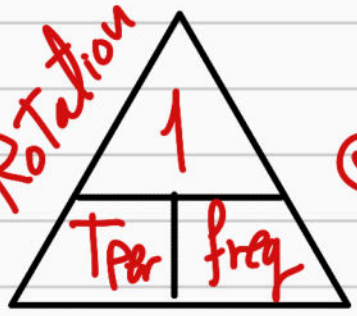


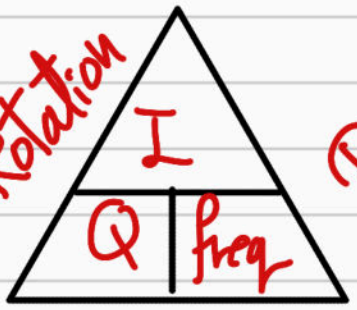


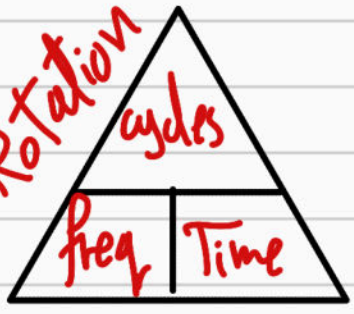


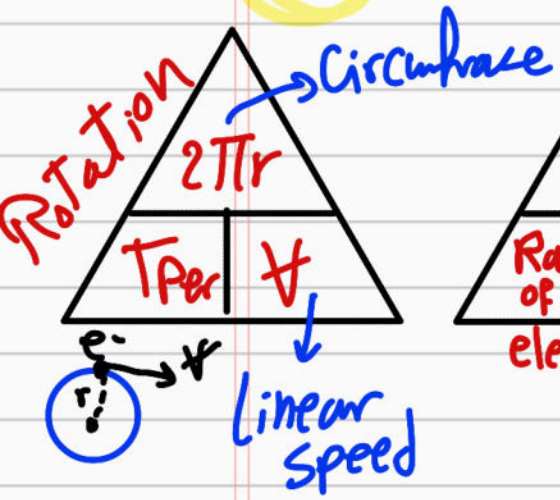


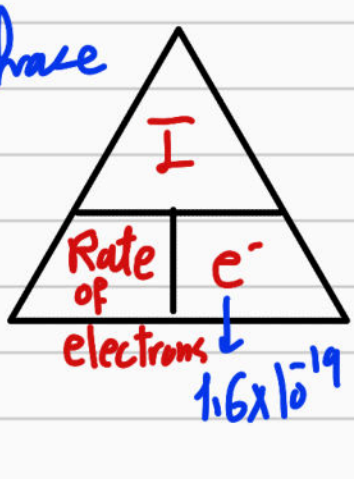


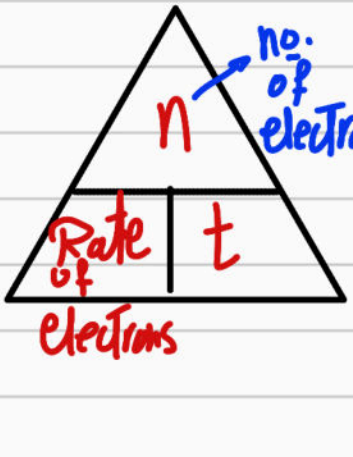


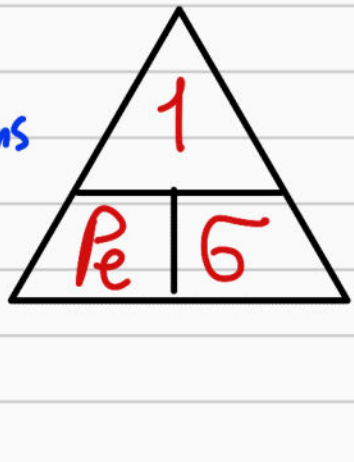




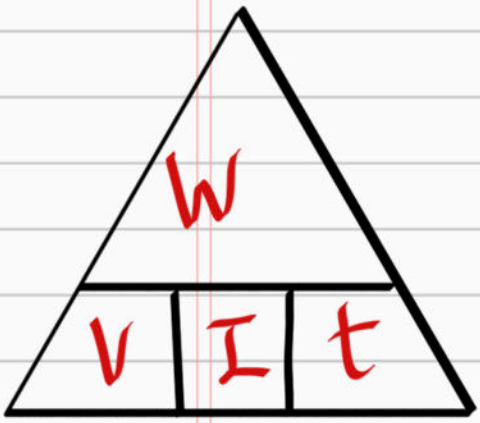


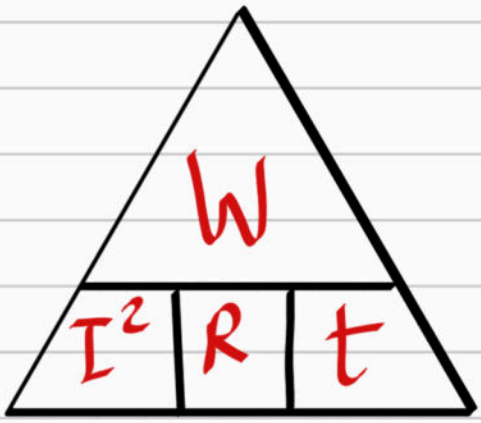


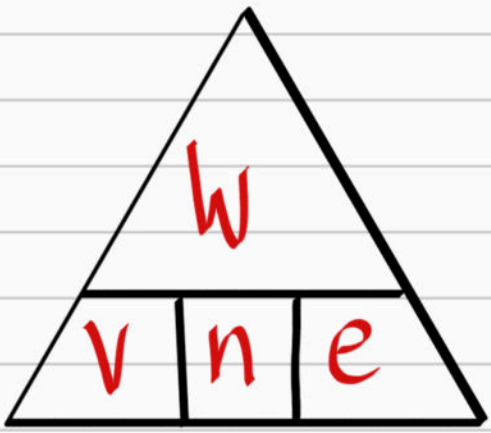




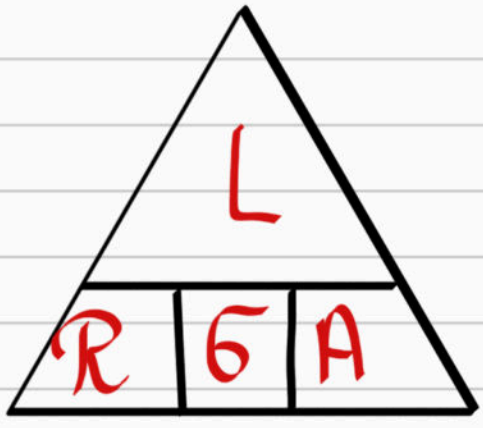
Rules Summary











P_e	L
R	A

n	e
I	t

V^2	t
W	R

Q	V
I	$2\pi r$

Rotation*

Rate energy
↳ Power

$P_{W_1} = 30\ W$

inc. To Quadrable → x4
inc. by Triple
inc. To 400...%
inc. by 300....%

$P_{W_2} = 120\ W$

Lesson ②

1. Seires & Parallel
2. Points Method → Redrawing
3. When To cancel a Resistance
4. Current distribution
5. Voltage distribution
6. Switches → Before & After
7. Rehostate



Seires



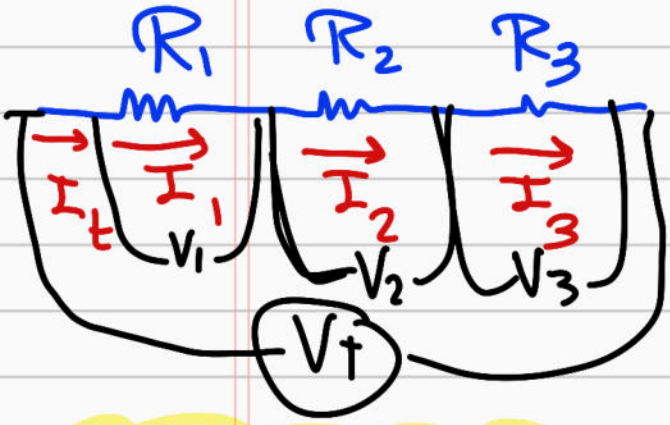
* كلهم على نفس الـ branch
* مفيش بينهم Nodes
* كلهم يبتدي منطقة نفس الـ I

Parallel



* لازم نفس الـ start وكمان نفس الـ end
* لازم كلهم نفس الـ (V)
* مفيش سلك في Battery

Seires ↑ $R_{eq} > R_{أكبر}$



$$I_t = I_1 = I_2 = I_3$$

$$V_t = V_1 + V_2 + V_3$$

$$R_{eq} = R_1 + R_2 + R_3$$

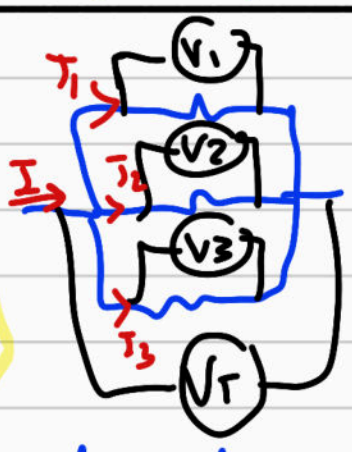
if identical

$$R_{eq} = n \times R$$

Parallel ↓ $R_{eq} < R_{أصغر}$

$$V_t = V_1 = V_2 = V_3$$

$$I_t = I_1 + I_2 + I_3$$



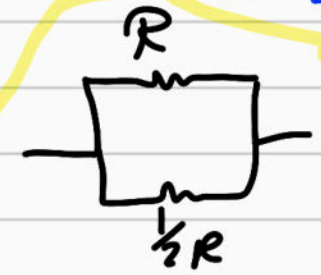
$$R_{eq} = \left[\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right]^{-1}$$

if Two only

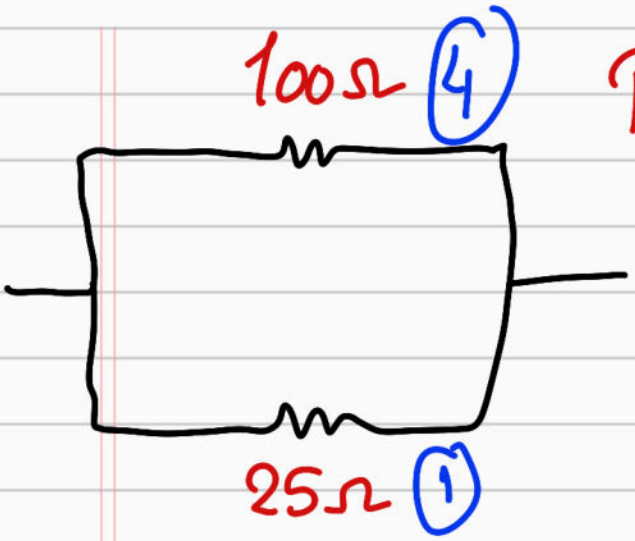
$$R_{eq} = \frac{R_1 \times R_2}{R_1 + R_2}$$

if identical

$$R_{eq} = R/n$$

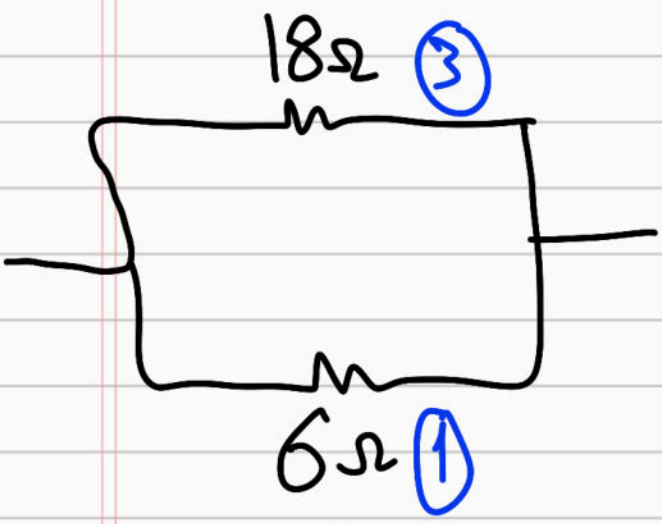


$$R_{eq} = R/3$$



$$R_{eq} = \frac{100 \times 25}{100+25}$$

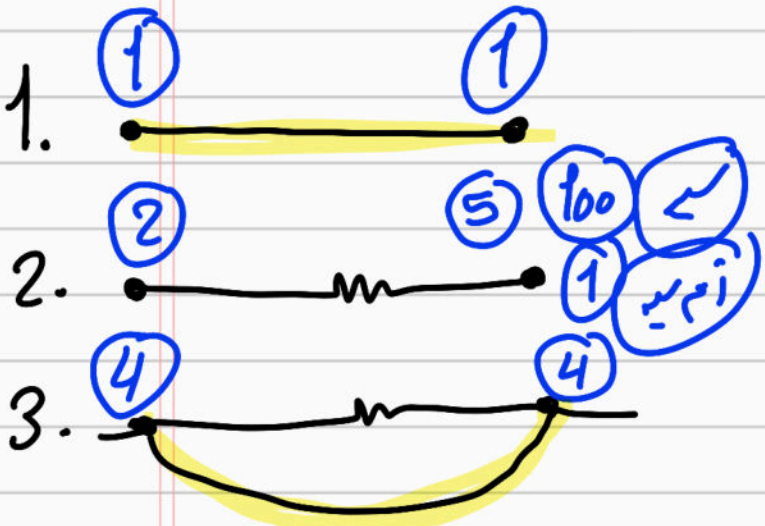
$$R_{eq} = \frac{100}{4+1} = \frac{100}{5} = 20\Omega$$

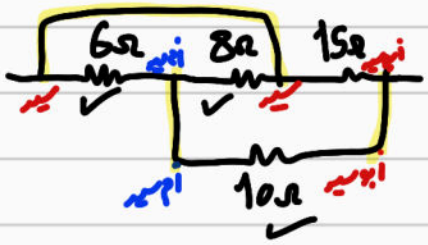


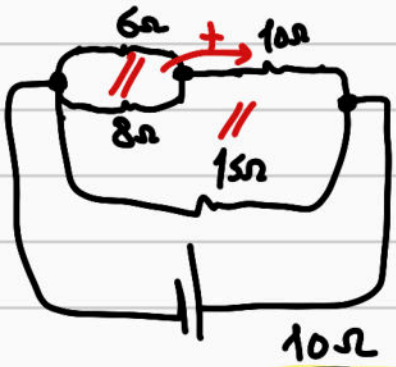
$$R_{eq} = \frac{18 \times 6}{18+6}$$

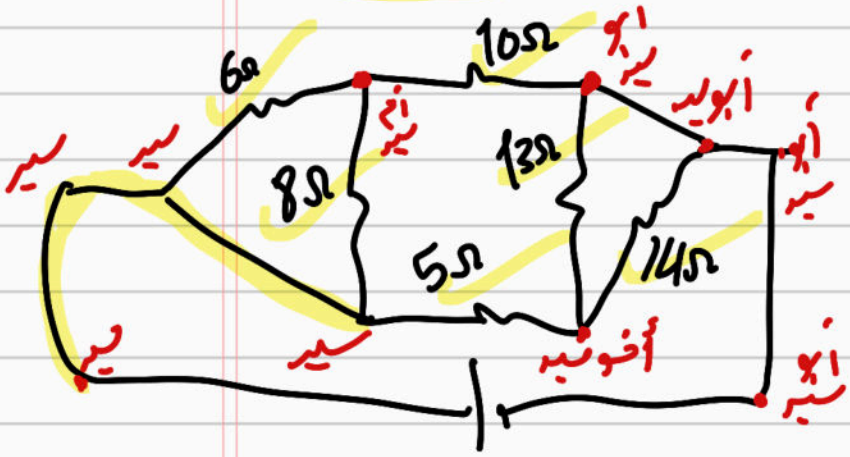
$$R_{eq} = \frac{18}{4} = 4.5\Omega$$

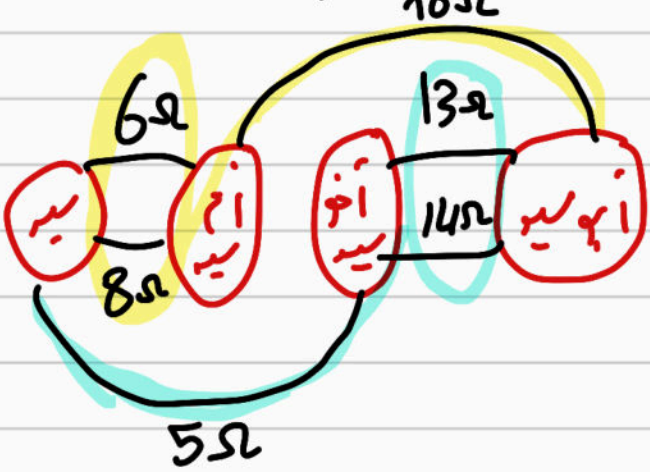
#Points Method:



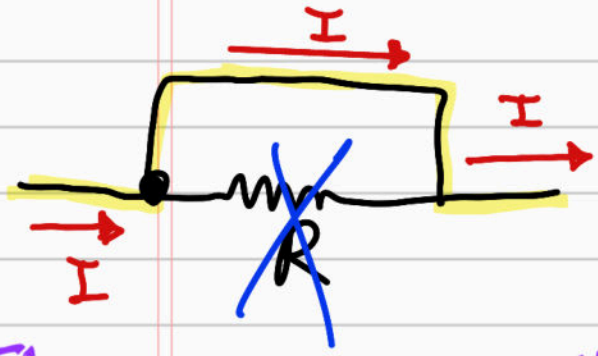




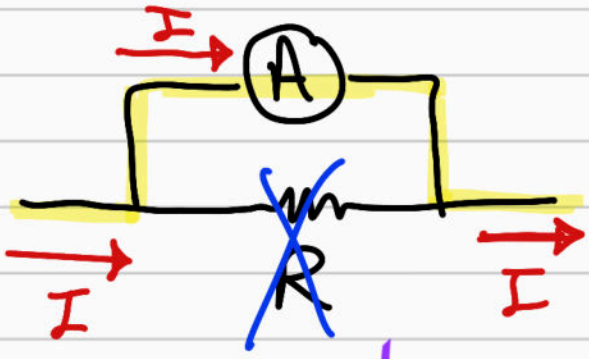




1 Short Circuit [Jumper Wire]:



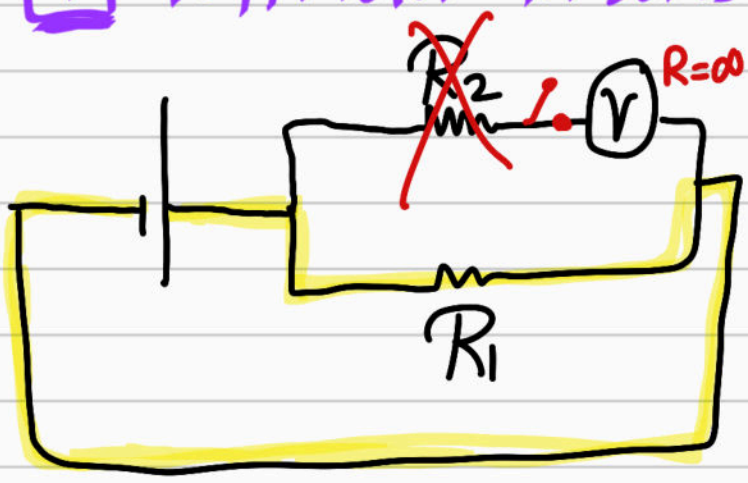
2 Ammeter in Parallel:



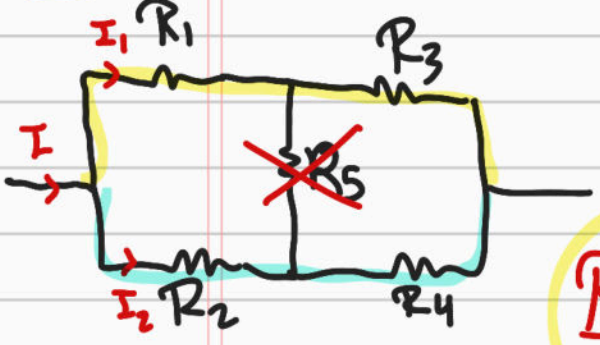
3 Open Switch



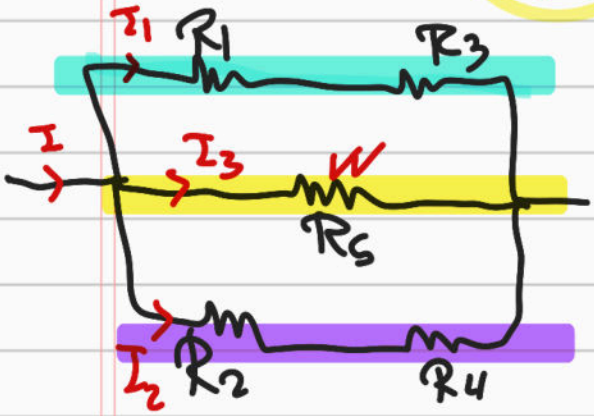
4 Voltmeter in Series



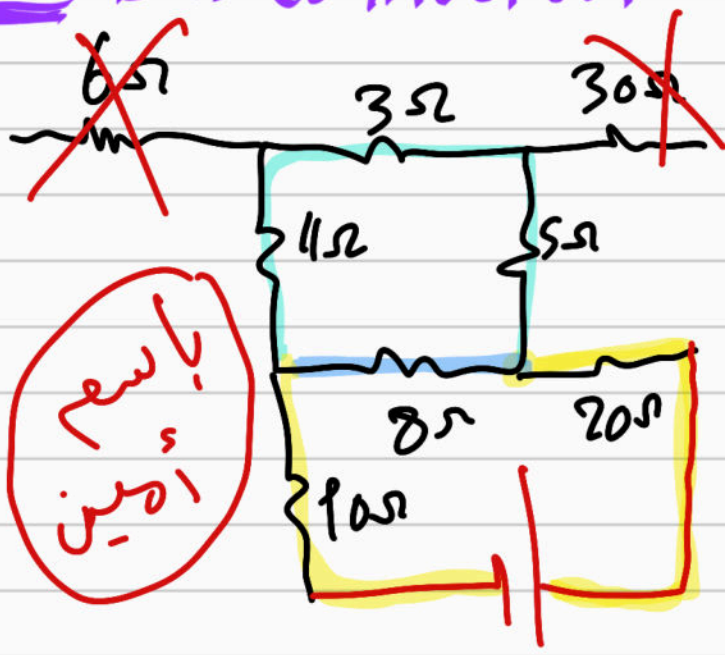
5 Winston Bridge:



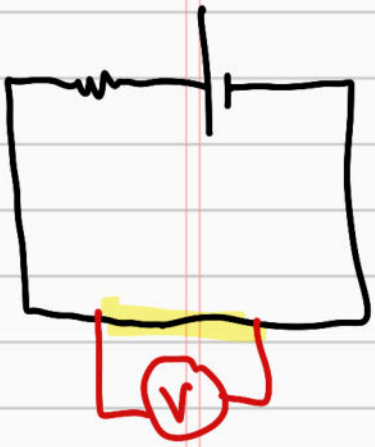
$$\frac{R_1}{R_2} = \frac{R_3}{R_4}$$

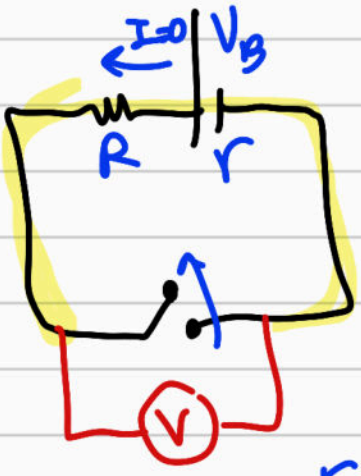


6 Dis connected:



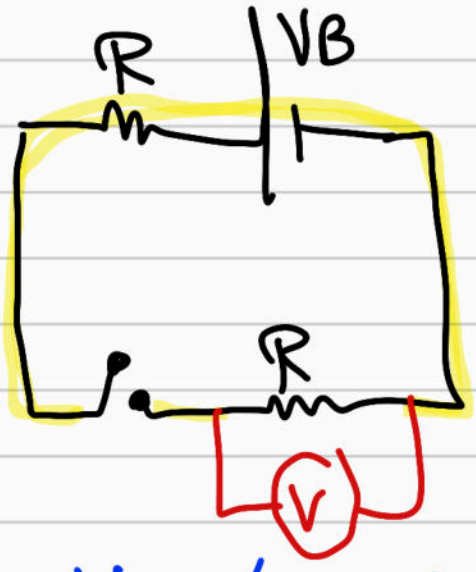
أبسط شي





$V = V_B - I[R+r]$ ($I \to 0$)

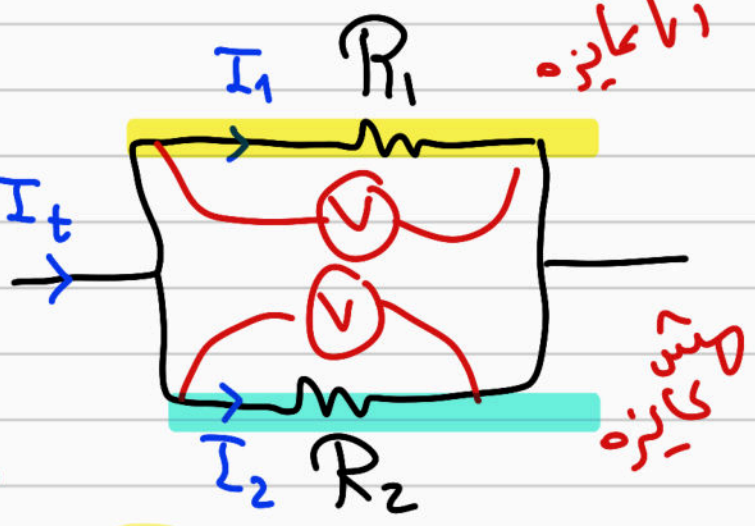
$V = V_B$



$V = IR = 0$ ($I \to 0$)

*Current distribution:

$$I_1 = \frac{I_t \times R_2}{R_1 + R_2} \rightarrow \text{C.D.}$$



$$R_{eq} = \frac{R_1 \times R_2}{R_1 + R_2}$$

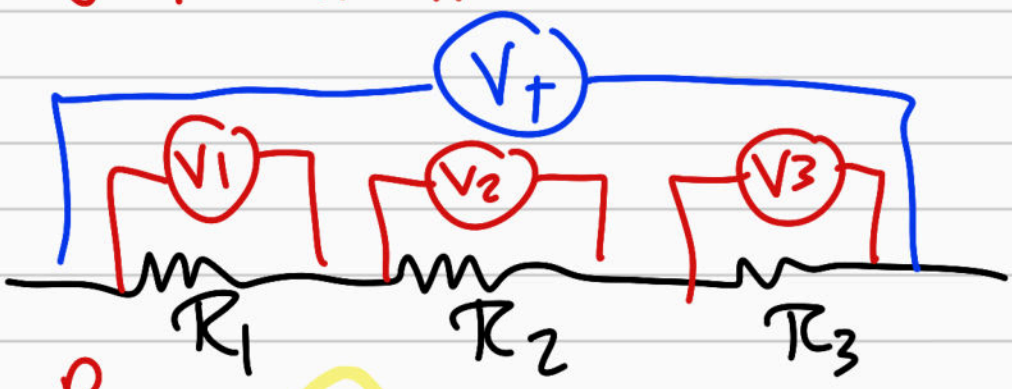
$$V = I_t \times R_{eq}$$

$$I_1 = \frac{V}{R_1} \qquad I_2 = \frac{V}{R_2}$$

$$I_1 : I_2 : I_t$$

$$R_2 : R_1 : R_1 + R_2$$

*Voltage disTribution:



$$V_1 = \frac{V_t \times R_1}{R_1 + R_2 + R_3}$$

$V_1 ; V_2 ; V_3 ; V_t$

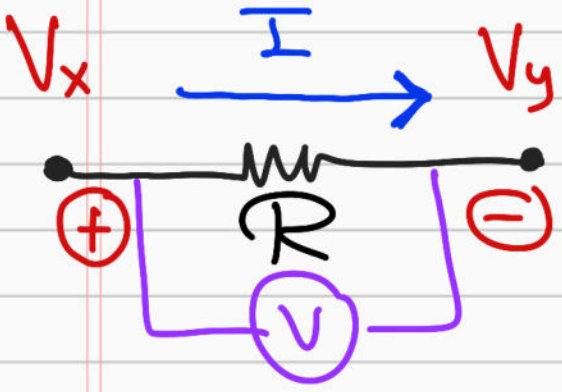
$R_1 ; R_2 ; R_3 ; R_1 + R_2 + R_3$

$$R_{eq} = R_1 + R_2 + R_3$$

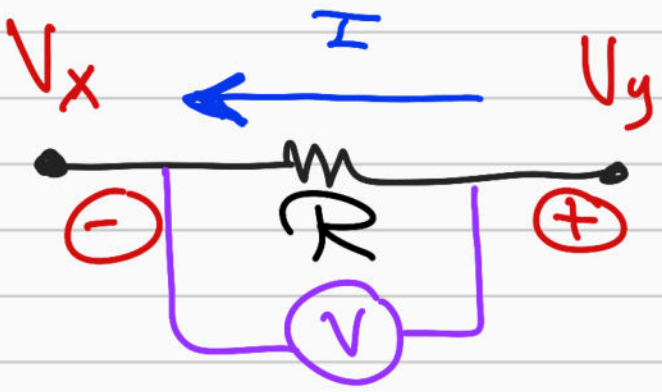
$$I = \frac{V_t}{R_{eq}}$$

$$V_1 = I \times R_1$$

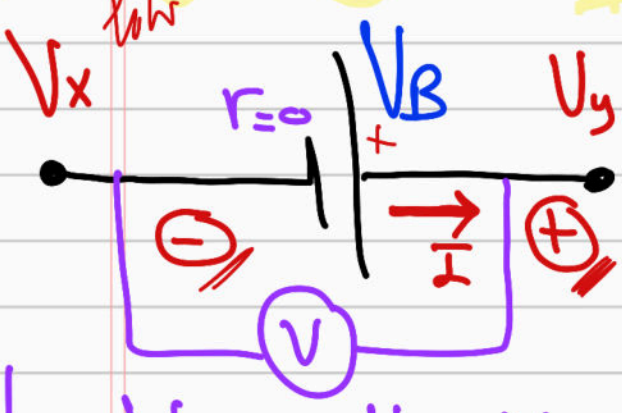
Potential difference between Two Points



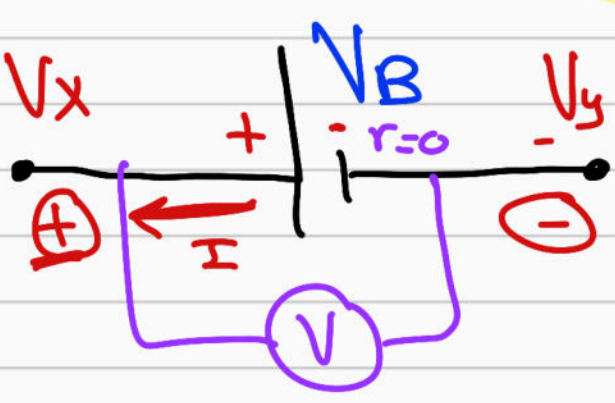
$V = V_x - V_y = IR$



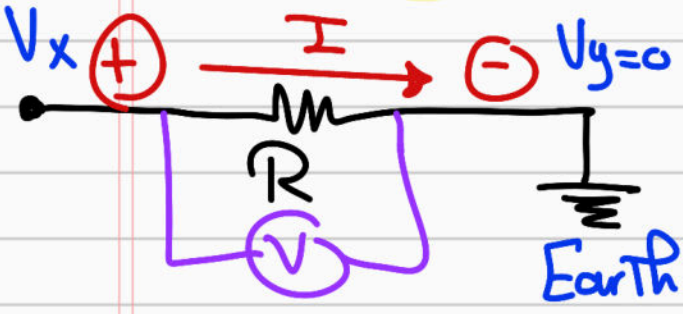
$V = V_y - V_x = IR$



$V = V_B = V_y - V_x$

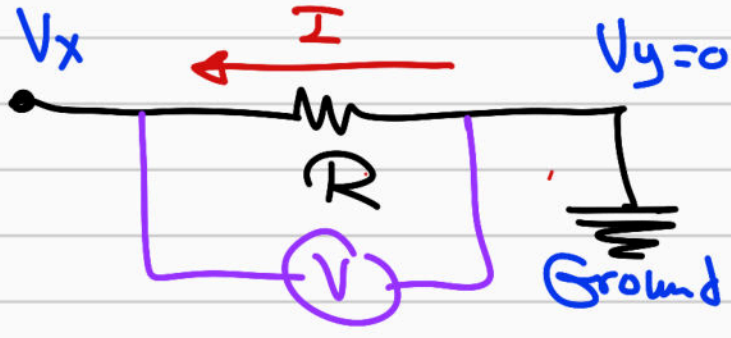


$V = V_B = V_x - V_y$



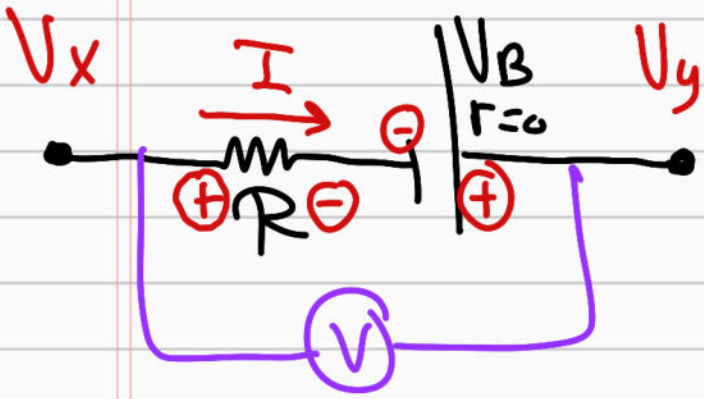
$V = V_x - V_y = IR$

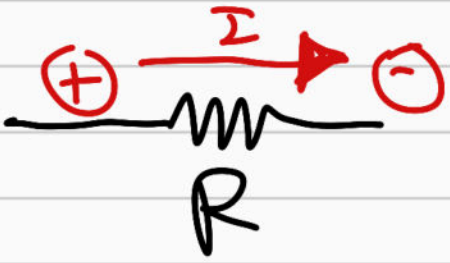
$V_y \to 0$



$V = V_y - V_x = IR$

$V_y \to 0$



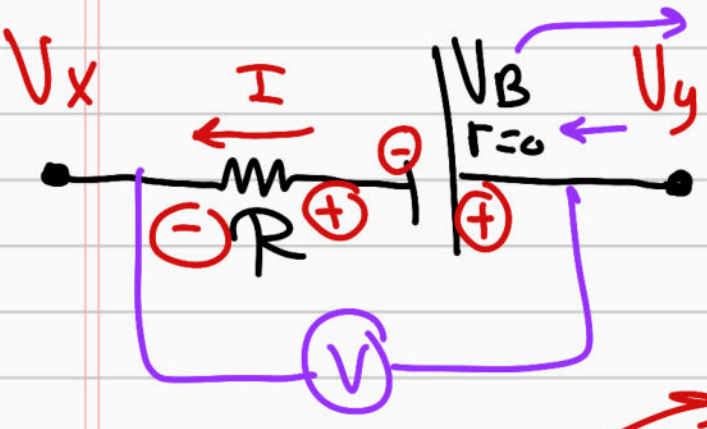




✓ $V = +IR - V_B = V_x - V_y$

OR

✓ $V = V_B \ominus IR = V_y - V_x$ (Discharging)

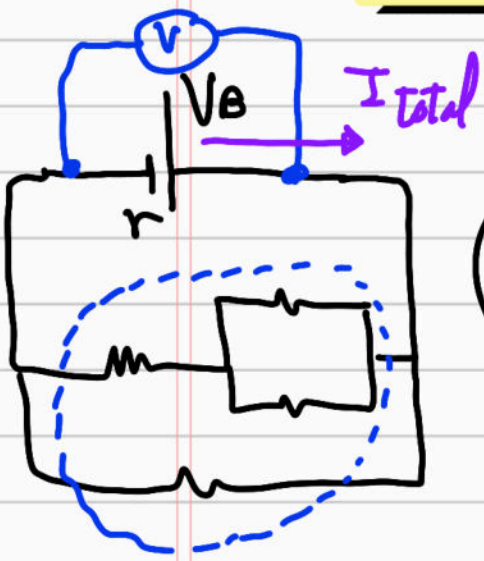


Charging

$V = V_B \oplus IR = V_y - V_x$ (Charging)

Chapter (1) Revision Part (2)

Lesson (3) Ohm's Law for closed circuit



$$I_{main/total} = \frac{V_B}{R_{eq}+r} = \frac{V}{R_{eq}} = \frac{V_{drop}}{r}$$

total — external — internal

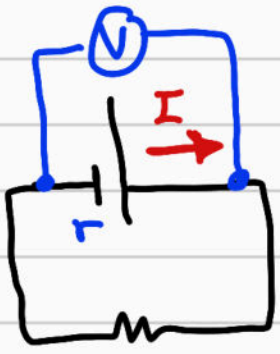
external — internal — external — internal

R_{eq}

$$V_B = I_{total} \times R_{total} = I_{main}[R_{eq}+r] = V + V_{drop}$$

$$V_{Terminals} = V = I_t \times R_{eq} = V_B - V_{drop}$$ Discharging

$$V_{drop} = I_t \times r = V_B - V_{Terminal}$$ Discharging

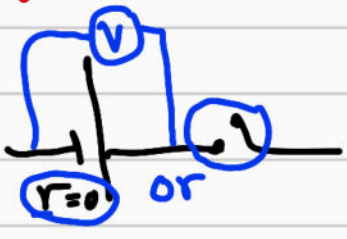


$V_{Terminal} < V_B \rightarrow$ Discharging $r \neq 0$

- $V_B = V + V_{drop}$
- $V = V_B - V_{drop}$
- $V_{drop} = V_B - V$

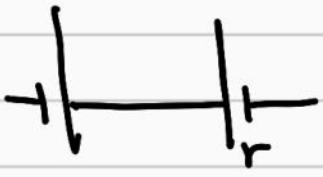
$V_{Terminal} = V_B \rightarrow$ $r = 0$, $I = 0$

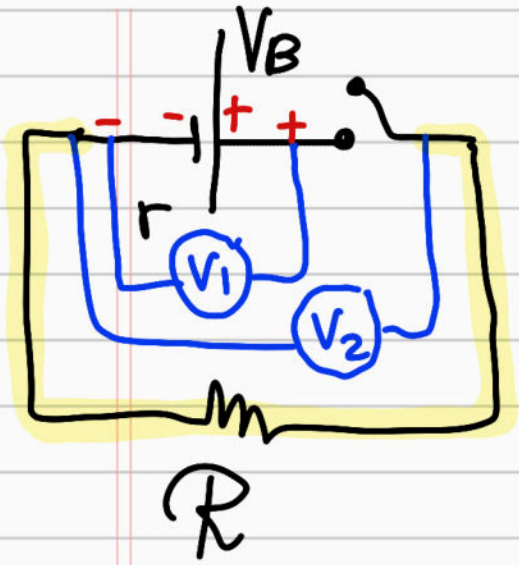
- $V_B = V$
- $V = V_B$
- $V_{drop} = 0$



$V_{Terminal} > V_B \rightarrow$ charging $r \neq 0$

- $V_B = V - V_{drop}$
- $V = V_B + V_{drop}$
- $V_{drop} = V - V_B$

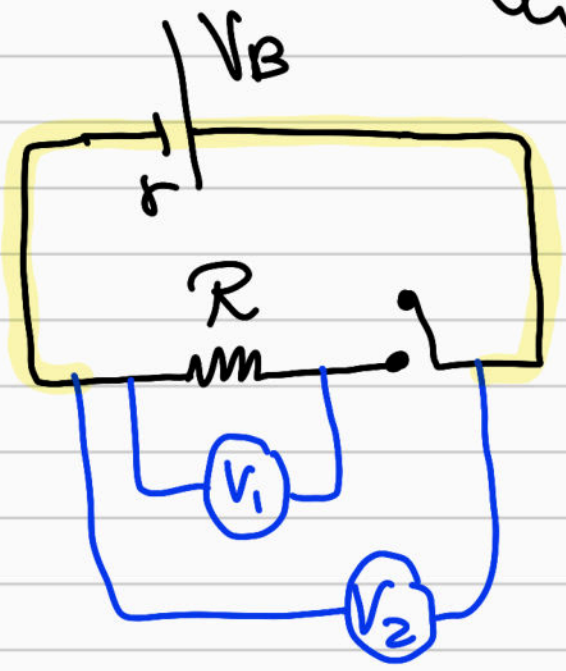




$V_1 = V_B$

$V_2 = Zero$

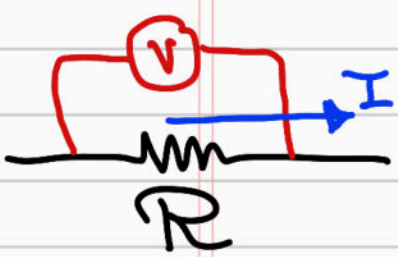
In Theory



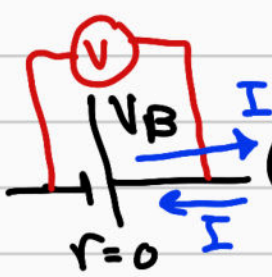
$V_1 = Zero$

$V_2 = V_B$

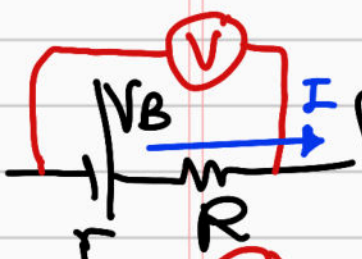
"Voltmeter Reading" Cases



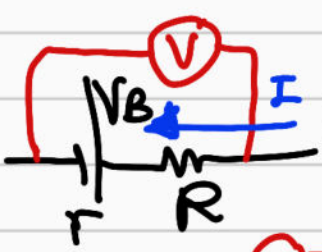
$V = IR$



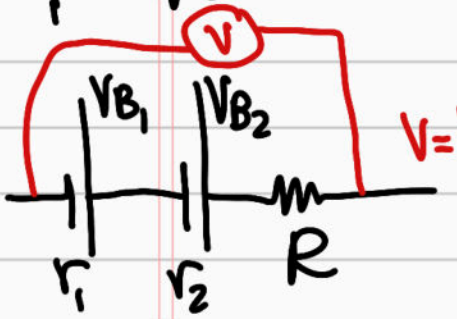
$V = V_B$



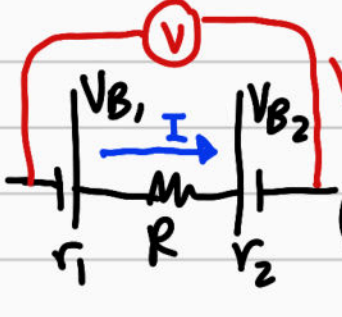
$V = V_B - I[R+r]$



$V = V_B + I[R+r]$



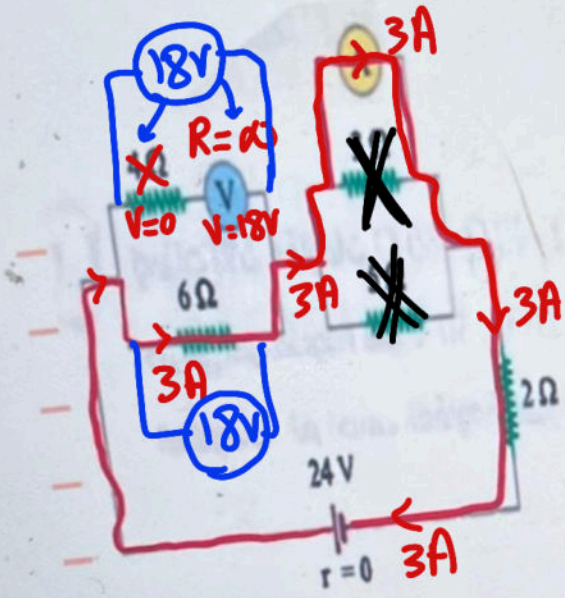
$V = V_{B_1} + V_{B_2}$ (+ charging, − discharging) $I[R+r_1+r_2]$



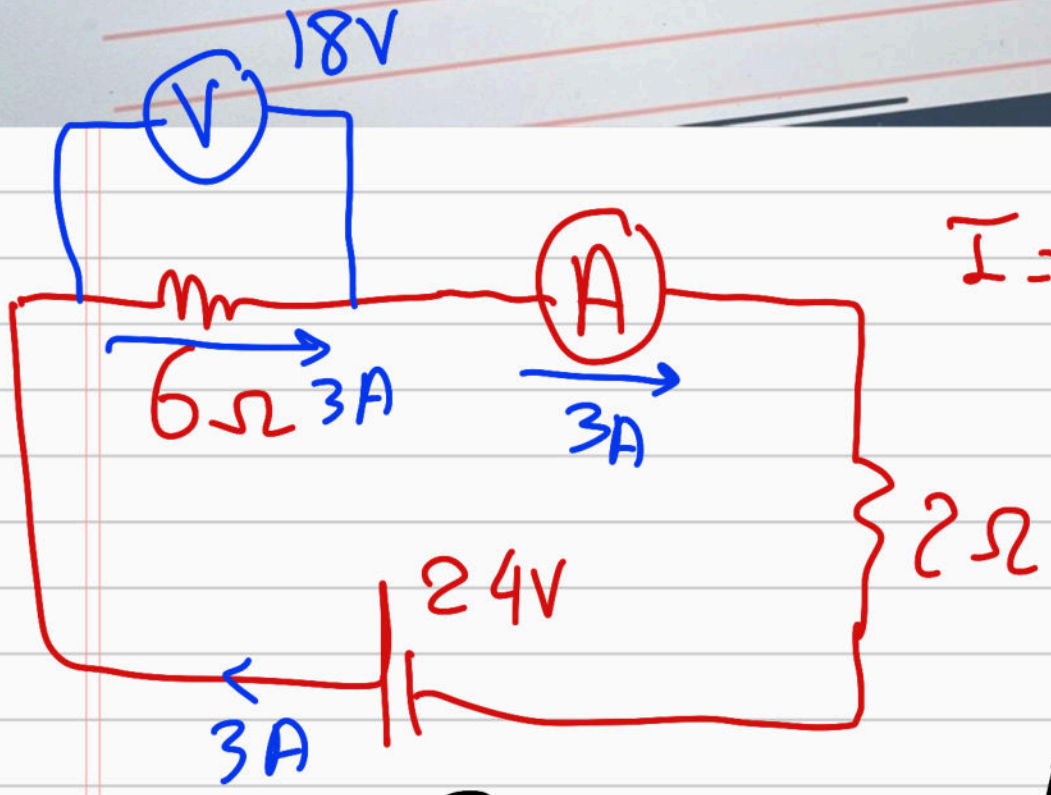
$\vec{V} = V_{B_1} - V_{B_2} - I[R+r_1+r_2]$

$V_{B_1} > V_{B_2}$

Reading of A- voltmeter

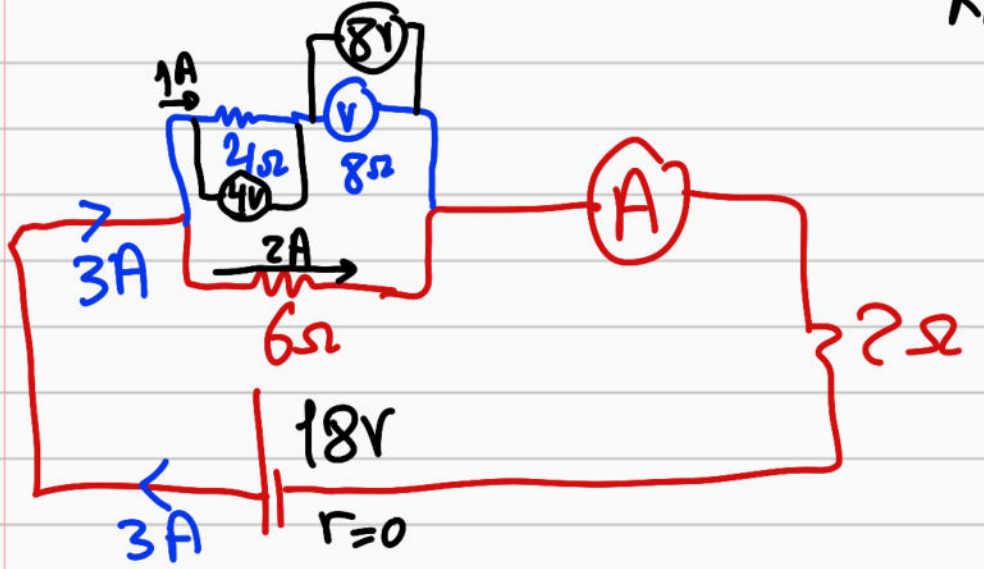


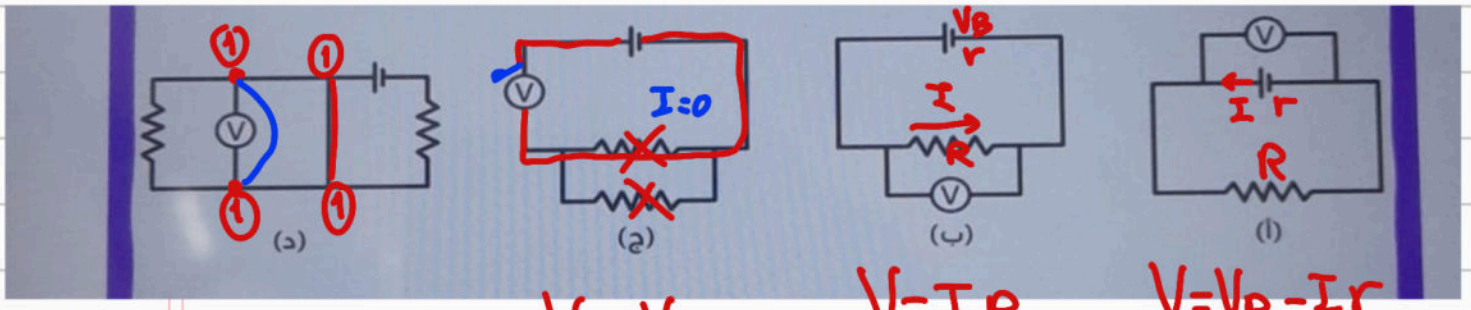
A	V
3A	0
6A	22V
1.2A	16V
3A	18V



$$I = \frac{24}{8} = 3A$$

$$R_{eq} = 4 + 2 = 6\Omega$$





$V = V_B$

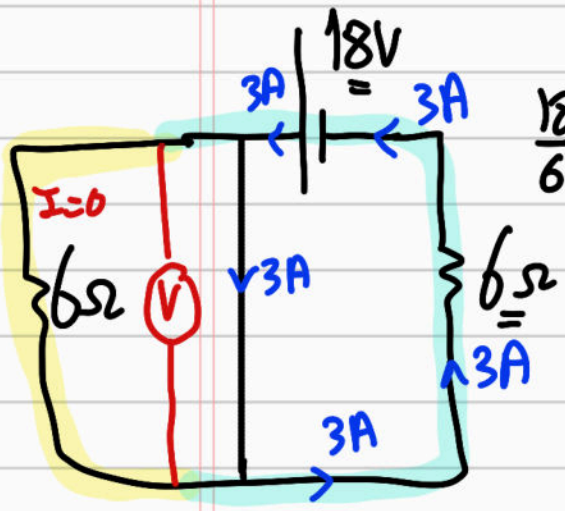
$V = IR$

$V = V_B - Ir$

$V = V_B - Ir$

$V = V_B$

$V = IR$

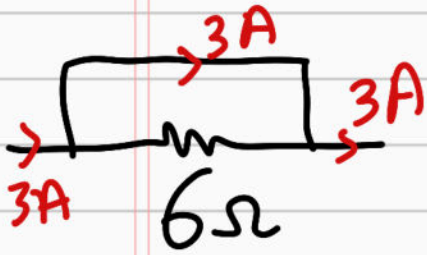


$\frac{18}{6} = 3A$

$V = 0 \times 6 = \text{Zero}$

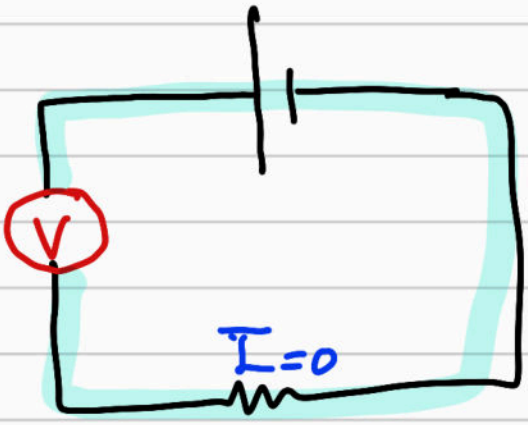
$V = V_B - IR$

$= 18 - 3 \times 6 = \text{Zero}$



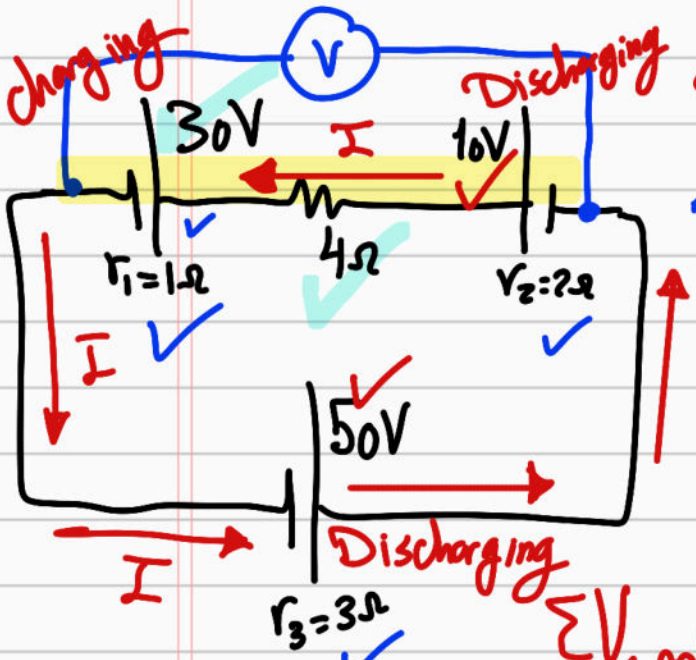
$V = V_B - IR/2$

$\rightarrow 0$



$R/2$

$10 - 3(2)$



Supply — Consume

$\text{Diff} = V = \frac{10 - 3(2)}{4} - \frac{30 + 3(1) + 3 \times 4}{45} = -41V$

$\text{Diff} = V = 50 - 3 \times 3 = 41V$

$I = \frac{50 + 10 - 30}{4 + 3 + 2 + 1} = 3A$

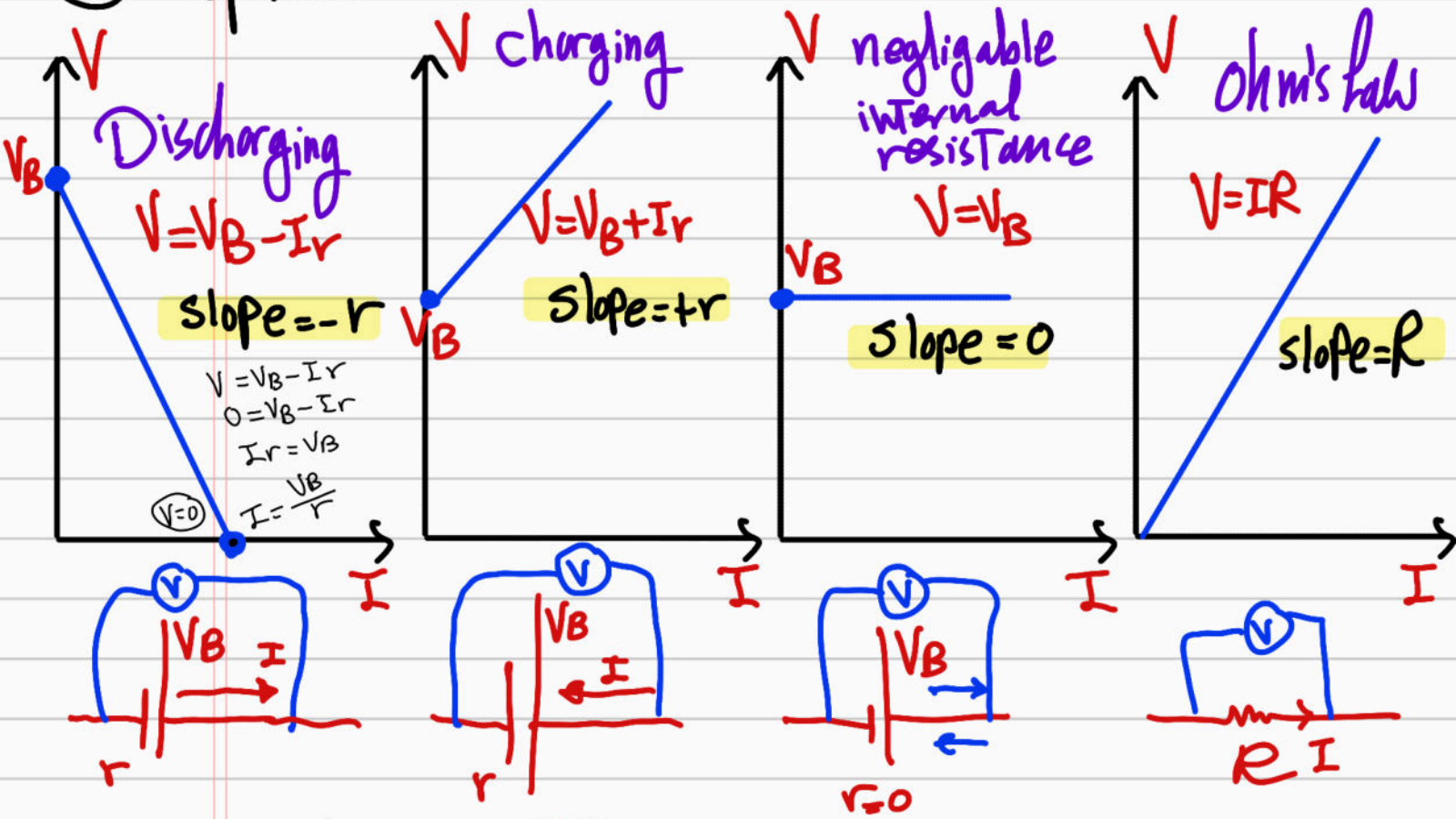
$\Sigma V_{\text{supply}} = 4 + 41 = 45V = \Sigma V_{\text{consume}}$

$$\Sigma V_{supply} = 50 + 10 = 60V$$

$$\Sigma V_{consume} = 30 + 3(4) + 3(3) + 3(2) + 3(1)$$

$$= 60V$$

Graphs:



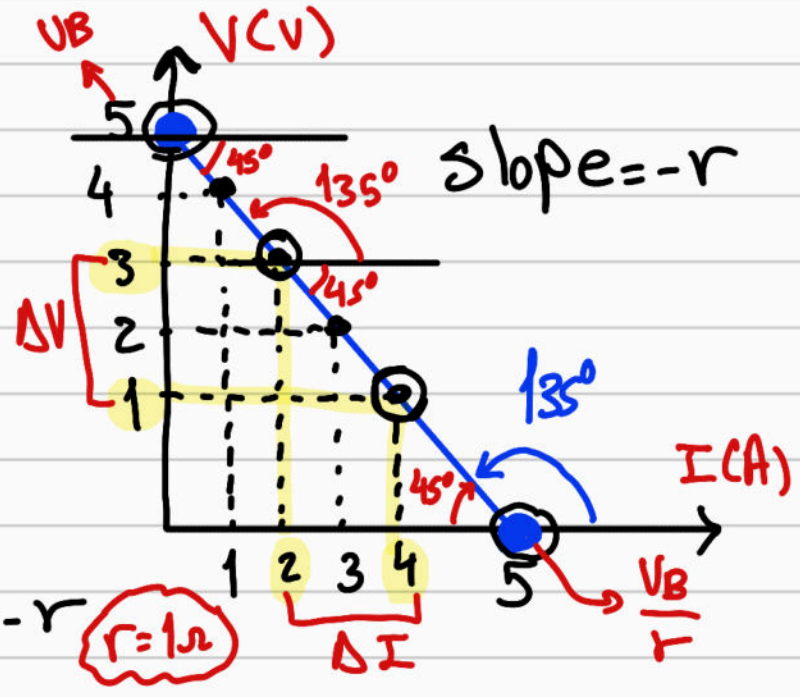
$$-r = -1 \rightarrow r = 1\Omega$$

$$slope = \frac{3-1}{2-4} = -1$$

$$slope = \frac{5-0}{0-5} = -1$$

$$slope = \tan(135) = -1$$

$$slope = \tan(-45°) = -1 = -r$$

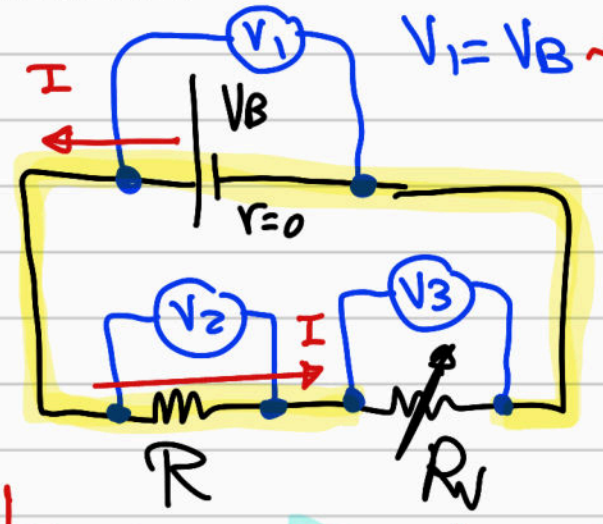


"Rehostate" → Variable Resistance *

Given:

$R_V \uparrow \quad R_{eq} \uparrow$

$I_{main} \downarrow \quad I_{R_V} \downarrow$

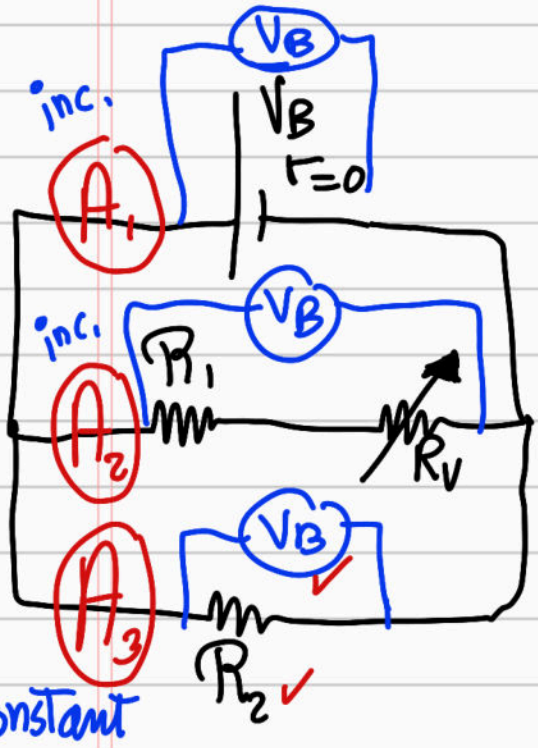


$V_1 = V_B \rightarrow$ const.

$V_2 = I_m \times R$

$V_3 = I_m \times R_V$

$V_3 = V_B - IR$

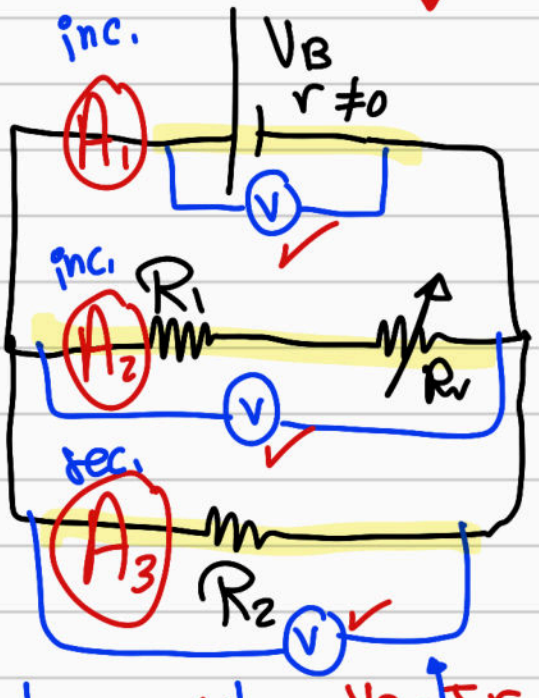


$R_V \downarrow$

$R_{eq} \downarrow$

$I_{main} \uparrow$

$I_{R_V} \uparrow$



$I_3 = \frac{V}{R_{2//}} = \frac{V_B - I_m r}{R_{2//}}$

Lesson ④

Kirchoff Laws

Kirchoff

Ohm

Kirchoff Current LaW "KCL"

Law of Conservation of electric Q Charge

$\sum I_{in} = \sum I_{out}$

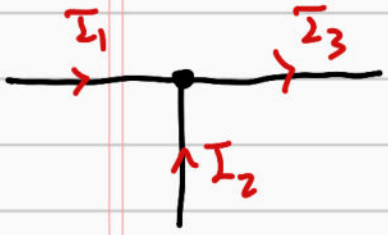
$\sum I_{node} = Zero$

Kirchoff Voltage LaW "KVL"

Law of Conservation of Energy

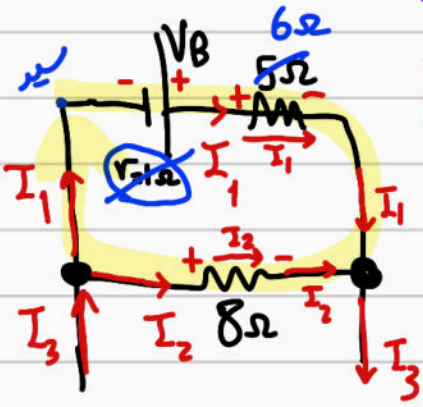
$\sum V_B = \sum IR$

$\sum V_{closed\ loop} = Zero$



$I_1 + I_2 - I_3 = 0$ ✓

$-I_1 - I_2 + I_3 = 0$ ✓



KVL:

$-V_B + 6I_1 - 8I_2 = 0$

$6I_1 - 8I_2 = V_B$

Mode ⑤ ②

I_1	I_2	I_3	V_B
1	1	-1	0
5	-7	0	-10
0	-7	-10	-80

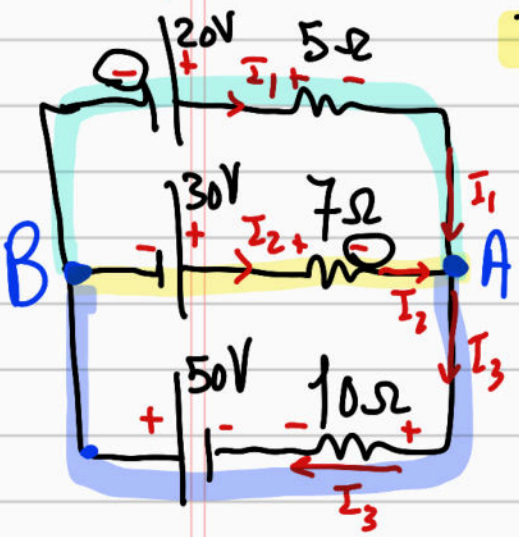
$ax + by + cz = d$

$x \rightarrow I_1 = \frac{78}{31} A$

$y \rightarrow I_2 = \frac{100}{31} A$

$z \rightarrow I_3 = \frac{178}{31} A$

Example



KCL:

$I_1 + I_2 - I_3 = 0$

KVL:

$-20 + 5I_1 - 7I_2 + 30 = 0$

$5I_1 - 7I_2 + 0I_3 = -10$

KVL: $+50 - 10I_3 - 7I_2 + 30 = 0$

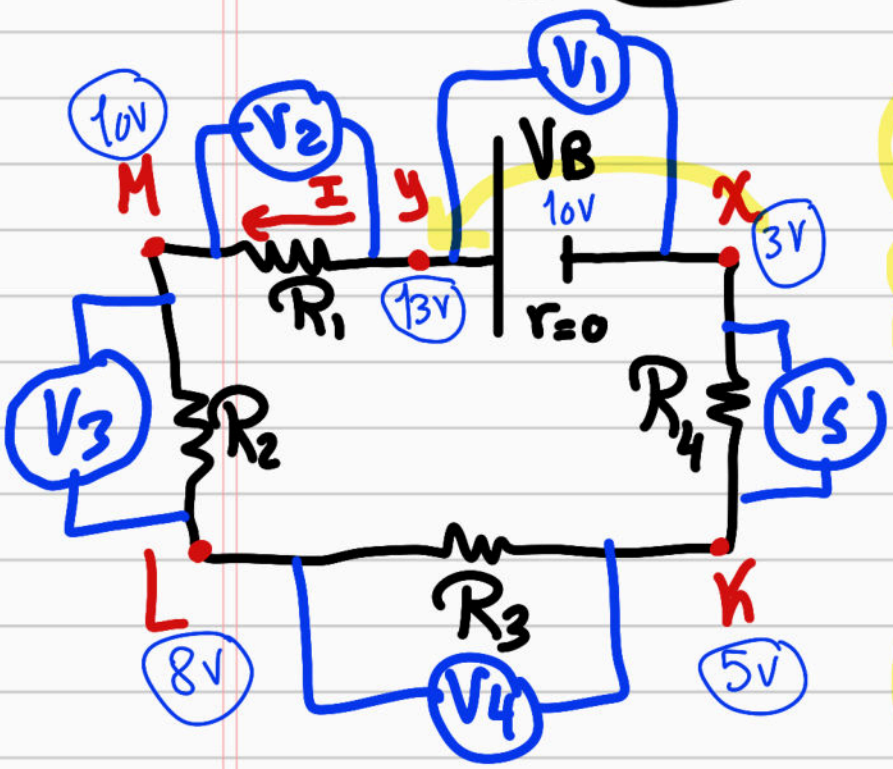
$0I_1 - 7I_2 - 10I_3 = -80$

$$V_{AB} = -5\left[\frac{78}{31}\right] + 20 = \frac{230}{31}\ V\ \#$$

$$V_{AB} = -7\left[\frac{100}{31}\right] + 30 = \frac{230}{31}\ V\ \#$$

$$V_{AB} = +10\left[\frac{178}{31}\right] - 50 = \frac{230}{31}\ V\ \#$$

Notes



$$V_1 = V_y - V_x = V_B$$

$$V_2 = V_y - V_M = IR_1$$

$$V_3 = V_M - V_L = IR_2$$

$$V_4 = V_L - V_K = IR_3$$

$$V_5 = V_K - V_x = IR_4$$